

Explainable Vision-Language Diagnostic Assistant for Chest Radiographs with Offline Rural Deployment

Mini Hackathon 1 Report — Design Phase

NLP Specialization Capstone Project

Section-07

Group-04

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1. System Architecture

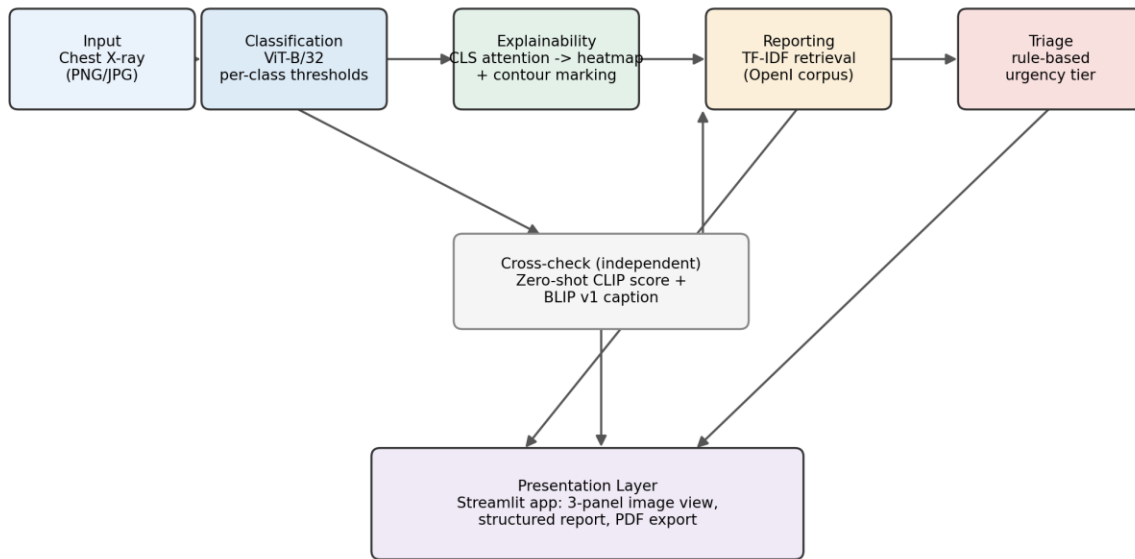
The system is organized as a linear pipeline with two independent output branches (vision explainability and text reporting) that recombine at the presentation layer. Every stage was chosen to keep the pipeline honest about what a CPU-only, small-data project can actually deliver, rather than overstating capability.

- **Input layer:** an uploaded chest X-ray image (PNG/JPG) enters the pipeline through the Streamlit front end.
- **Classification layer:** a fine-tuned ViT-B/32 model outputs a probability per condition across 11 classes; each probability is compared against its own tuned threshold (not one global cutoff), producing a list of flagged conditions.
- **Explainability layer:** for the primary (highest-confidence) flagged condition, CLS-token attention weights from the final transformer block are extracted to produce a heatmap and a red-contour region marking.
- **Reporting layer:** for every flagged condition, a TF-IDF retrieval index returns the closest-matching real report text from the OpenI corpus; a combined, multi-condition report is assembled with precautions and a triage recommendation.
- **Triage layer:** a deterministic severity table combines all flagged conditions' severities and confidence levels into a single urgency tier.
- **Cross-check layer:** zero-shot CLIP scoring and a BLIP v1 caption are computed independently, from architecturally distinct pretrained models, and displayed as supplementary signals only.
- **Presentation layer:** a Streamlit web app displays three image views (original, heatmap, marked region), the structured report, and a downloadable PDF combining all outputs plus patient details and a technical addendum.

2. Flow Diagram

The diagram below traces a single image from upload through to the final clinician-facing output. The classification and explainability stages feed the cross-check branch, which contributes a secondary opinion into both the report and the presentation layer, while the reporting and triage stages feed the presentation layer directly.

End-to-End Pipeline: Explainable Vision-Language Diagnostic Assistant



3. System Management and Compatibility

The system is deliberately built from CPU-friendly, widely available components so it can be managed and deployed without specialist infrastructure:

- **Runtime footprint:** all models (frozen-backbone ViT-B/32, DenseNet121 baseline, CLIP, BLIP v1) run on CPU; no GPU or cloud inference service is required at run time.
- **Modular files:** each pipeline stage lives in its own module (model.py, train.py, tune_thresholds.py, gradcam.py, report_index.py, report_generator.py, triage.py, clip_verify.py, blip_caption.py, report_pdf.py), so a stage can be upgraded or replaced without touching the others.
- **Compatibility:** built on PyTorch/torchvision, HuggingFace Transformers, scikit-learn, OpenCV, pandas, Streamlit, and fpdf2 — all standard, actively maintained Python libraries with broad OS support (Windows/Linux/macOS).
- **Data management:** the NIH ChestX-ray14 sample and the OpenI report corpus are acquired via Kaggle/kagglehub and cached locally, so re-runs do not require repeated downloads.
- **Extensibility:** the retrieval index and threshold tables are external artifacts (not hard-coded), so the class list, report corpus, or thresholds can be updated without retraining the classifier.

4. Previous Models

Two backbones were trained and evaluated on the same 11-class, held-out validation split (840 images) before settling on the final design:

Backbone	Classes	Macro F1 (0.5 threshold)	Macro F1 (tuned thresholds)
DenseNet121 (CNN)	11	0.254	0.304
ViT-B/32 (Transformer)	11	0.230	0.260

DenseNet121 was the initial CNN baseline used to benchmark the later ViT swap. It scored marginally higher on raw macro F1, which is consistent with a well-documented property of Vision Transformers: unlike CNNs, ViT lacks built-in locality and translation-invariance biases and typically needs more training data to reach comparable performance — an expected outcome on a ~4,800-image training set, not an implementation defect. ViT-B/32 was retained as the primary model because it also unlocked attention-based explainability, which the project's interpretability goals depend on.

5. New Development and Novelty

- **Explainability method change:** gradient-based Grad-CAM, adapted from CNNs to the ViT's final transformer block, produced degenerate output (zero variance across patches) due to token homogenization from limited fine-tuning. This was diagnosed and replaced with CLS-token attention visualization, the standard ViT-appropriate technique, restoring real spatial variation (heatmap standard deviation 0.277 vs. 0.000).
- **Retrieval-augmented, not generative, reporting:** report language is retrieved from real radiology reports via TF-IDF similarity rather than generated freely by a language model, trading open-endedness for grounded, non-fabricated clinical phrasing.
- **Post-hoc per-class threshold tuning:** decision thresholds are optimized per condition against the validation precision-recall curve after training, improving macro F1 and, critically, recall on higher-severity conditions without any additional training.
- **Independent cross-checks:** zero-shot CLIP scoring and a BLIP v1 caption are layered on top of the primary pipeline as architecturally distinct, non-fine-tuned second opinions, clearly separated from the diagnostic verdict.
- **Multi-condition reporting:** the reporting stage was upgraded to flag and report on every condition above its threshold, not just the single top prediction, better matching real multi-finding chest X-rays.

6. Can Be Executed in Real Life

The design was validated qualitatively as well as quantitatively. On a held-out example with ground truth Cardiomegaly, Edema, and Effusion, the pipeline correctly flagged Cardiomegaly at 99.9% confidence and, after the multi-condition reporting upgrade, correctly flagged Effusion as well. Edema was not flagged because it was one of three conditions excluded from the class list for insufficient training examples (118 positive instances) — a documented, deliberate scope limitation rather than a model failure.

Feasibility for real-world execution rests on choices made throughout the design:

- Every model runs on CPU, so the assistant can be piloted on ordinary clinic hardware rather than requiring a GPU server.
- The deliberate precision/recall trade-off (favoring recall) matches real triage priorities: a missed finding is more costly than a false alarm a clinician can quickly rule out.
- Outputs are explicitly framed as decision support (heatmap, retrieved report language, triage tier) rather than an autonomous diagnosis, keeping a clinician in the loop.
- Known limitations — a small training subset, low precision on some rare classes, and an approximate quadrant-based region description — are documented so a pilot deployment is entered with realistic expectations.

7. Offline Deployment

Offline, rural-clinic use was a first-class design constraint, not an afterthought:

- **No network dependency at inference:** once model weights, the retrieval index, and the report corpus are downloaded and cached, the Streamlit app runs entirely locally with no calls to external APIs or cloud services.
- **CPU-only footprint:** all four models used at run time (ViT-B/32 classifier, CLIP, BLIP v1, plus the TF-IDF retrieval index) were deliberately chosen or fine-tuned to be CPU-viable, avoiding GPU-dependent components such as BLIP-2.
- **Self-contained report generation:** the PDF export (fpdf2) is generated locally, so a clinician can leave the session with a printable record even without connectivity to email or a hospital information system.
- **Practical implication:** the whole pipeline can, in principle, run on a single offline laptop placed at a rural clinic, with periodic (not continuous) connectivity needed only to update models or the report corpus.

8. Use Cases

- Rural or under-resourced clinics without an on-site radiologist, where the assistant provides a first-pass read to support triage decisions.

- Screening/prioritization tool for community health workers to flag which chest X-rays most urgently need remote specialist review.
 - Second-opinion support for a general practitioner reviewing a chest X-ray outside their specialty, with retrieved report language and a heatmap as an aid, not a verdict.
 - Training and education aid, letting trainees compare the model's attention region and retrieved report language against their own reading of the film.
 - Research and iteration platform: the modular pipeline (swappable backbone, threshold table, retrieval corpus) makes it a base for further experimentation as more data or compute becomes available.
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